

Multiple Choice (10 marks)

1. If the total energy of a particle of mass m is equal to twice its rest energy, then the magnitude of the particle's relativistic momentum is
 - (a) $mc/2$
 - (b) $mc/(2^{1/2})$
 - (c) mc
 - (d) $(3^{1/2})mc$
2. De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?
 - (a) Planck's constant
 - (b) Boltzmann's constant
 - (c) The Rydberg constant
 - (d) The speed of light
3. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R . The time required for one revolution is
 - (a) independent of M
 - (b) proportional to $m^{1/2}$
 - (c) linear in R
 - (d) proportional to $R^{3/2}$
4. A refracting telescope consists of two converging lenses separated by 100 cm. The eyepiece lens has a focal length of 20 cm. The angular magnification of the telescope is
 - (a) 4
 - (b) 5
 - (c) 6
 - (d) 20
5. A resistor in a circuit dissipates energy at a rate of 1 W. If the voltage across the resistor is doubled, what will be the new rate of energy dissipation?
 - (a) 0.25 W
 - (b) 0.5 W
 - (c) 1 W
 - (d) 4 W

True / False (10 marks)

Answer True or False

6. True or False: In a reversible thermodynamic process, the temperature of the system must remain constant throughout the process.
7. True or False: Electromagnetic radiation emitted from a nucleus is often in the form of microwaves.
8. True or False: The speed of light in a nonmagnetic dielectric material with a dielectric constant of 4.0 is 3.0×10^8 m/s.
9. True or False: The Hall coefficient can be used to determine the sign of charge carriers in a doped semiconductor.
10. True or False: The negative muon, μ^- , *has properties most similar to those of the electron, including charge*

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the similarities between the emission spectrum of doubly ionized lithium (Li^{++}) and hydrogen, focusing on the wavelength changes and the underlying physics principles.
12. A tube of water is moving at $1/2 c$ relative to a lab frame. Analyze how the speed of a light beam, entering the tube in the same direction, is affected by the index of refraction of water.

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